

UNCERTAINTY

CHAPTER 14

Outline

- ◇ Uncertainty
- ◇ Probability
- ◇ Syntax
- ◇ Semantics
- ◇ Inference rules

Uncertainty

Let action $A_t =$ leave for airport t minutes before flight
Will A_t get me there on time?

Problems:

- 1) partial observability (road state, other drivers' plans, etc)
- 2) noisy sensors (KCBS traffic reports)
- 3) uncertainty in action outcomes (flat tire, etc.)
- 4) immense complexity of modelling and predicting traffic

Hence a purely logical approach either

- 1) risks falsehood: " A_{25} will get me there on time"
- or 2) leads to conclusions that are too weak for decision making
" A_{25} will get me there on time if there's no accident on
and it doesn't rain and my tires remain intact etc etc."

(A_{1440} might reasonably be said to get me there on time
but I'd have to stay overnight in the airport ...)

Methods for handling uncertainty

Default or nonmonotonic logic:

Assume my car does not have a flat tire

Assume A_{25} works unless contradicted by evidence

Issues: What assumptions are reasonable? How to handle contradiction?

Rules with fudge factors:

$A_{25} \mapsto_{0.3} \text{get there on time}$

$\text{Sprinkler} \mapsto_{0.99} \text{WetGrass}$

$\text{WetGrass} \mapsto_{0.7} \text{Rain}$

Issues: Problems with combination, e.g., *Sprinkler* causes *Rain*

Probability

Given the available evidence,

A_{25} will get me there on time with probability 0.04

Mahaviracarya (9th C.), Cardano (1565) theory of gambling

(Fuzzy logic handles *degree of truth* NOT uncertainty e.g.,

WetGrass is true to degree 0.2)

Probability

Probabilistic assertions *summarize* effects of

 laziness: failure to enumerate exceptions, qualifications, etc.

 ignorance: lack of relevant facts, initial conditions, etc.

Subjective or Bayesian probability:

Probabilities relate propositions to one's own state of knowledge

$$\text{e.g., } P(A_{25} | \text{no reported accidents}) = 0.06$$

These are *not* assertions about the world

Probabilities of propositions change with new evidence:

$$\text{e.g., } P(A_{25} | \text{no reported accidents, 5 a.m.}) = 0.15$$

(Analogous to logical entailment status $KB \models \alpha$, not truth.)

Making decisions under uncertainty

Suppose I believe the following:

$$P(A_{25} \text{ gets me there on time} | \dots) = 0.04$$

$$P(A_{90} \text{ gets me there on time} | \dots) = 0.70$$

$$P(A_{120} \text{ gets me there on time} | \dots) = 0.95$$

$$P(A_{1440} \text{ gets me there on time} | \dots) = 0.9999$$

Which action to choose?

Depends on my preferences for missing flight vs. airport cuisin

Utility theory is used to represent and infer preferences

Decision theory = utility theory + probability theory